METRIC

DOD-C-24594(SH)
AMENDMENT 1
1 September 1982

MILITARY SPECIFICATION

CELL, STORAGE, SILVER-ZINC ALKALINE TYPE

(FOR DEEP SUBMERGENCE APPLICATIONS) (METRIC)

This amendment forms a part of Military Specification DOD-C-24594(SH), dated 6 September 1979, and is approved for use by the Naval Sea Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

2.1, under "SPECIFICATIONS, FEDERAL": Delete reference to "QQ-B-613", "QQ-B-626", "QQ-B-637", "QQ-B-639", and "QQ-C-576".

Beneficial comments statement at bottom of page, line 3: Delete "SEA 3112" and substitute "SEA 55Z3".

PAGE 2

2.2: Add the following:

"AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
B 21 - Naval Brass Rod, Bar and Shapes. (DOD Adopted)
B 36 - Brass Plate, Sheet, Strip, and Rolled Bar. (DOD Adopted)
B 121 - Leaded Brass Plate, Sheet, Strip, and Rolled Bar. (DOD Adopted)
B 152 - Copper Sheet, Strip, Plate and Rolled Bar. (DOD Adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)"

PAGE 3

3.4.5.1, line 4: Delete "0.05" and substitute "0.076".

3.4.5.3, line 2: Delete "3.35" and substitute "3.46".

FSC 6140

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3.4.6: Delete and substitute:

"3.4.6 Terminals. That part of the terminal mating with the plug-in type intercell connector (see 3.4.6.1) shall be made of copper alloy No. 155 or equal. Other current carrying terminal parts shall be made of solid copper or copper alloys in accordance with ASTM B 21, B 36, B 121, or B 152. Terminal parts shall be silver-plated in accordance with QQ-S-365, type III, grade B, with a nickel undercoat in accordance with QQ-N-290, class 1, grade F, form SB. Assembly of the terminals shall be by mechanical attachment or brazing. Soldering shall not be used. Brazing shall be in accordance with MIL-B-7883, grade B, using brazing alloys in accordance with QQ-B-650 or QQ-B-654. Terminal posts shall be symmetrically placed with respect to an axis through the center of the cell top. One terminal post shall be provided for each polarity and shall be a part of a single, silver-plated copper strap or ferrule to which the plate tabs will be attached. Terminal posts shall be circular in cross section. Provisions shall be made to permit measuring the voltage drop between the terminal and the intercell connector when mated."

Add as new paragraph 3.4.6.1:

"3.4.6.1 Intercell connectors. Intercell connectors shall be in accordance with the contract or order. The completed plug-in intercell connector design shall be confirmed during the first design review. Provisions shall be made to permit measuring the voltage drop between the terminal and the intercell connector when mated."

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3.6.2: Delete and substitute:

"3.6.2 Installation and checkout parts. Installation and checkout parts shall be provided as specified (see 6.2.1) and shall include the following for each group of cells:

(a) Fifteen entrainment eliminators with O-rings and netting.
(b) Requisite electrolyte for all dry spare cell filling (to be provided in separate polyethylene carboys of 3 to 4 liters).
(c) Two electrolyte level indicators.
(d) Two electrolyte adjustment devices.
(e) Eight cell markers (blank).
(f) Two plug-in intercell connectors.
(g) Two plug-in cell jumper cables."

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4.5.3.5: Delete and substitute:

"4.5.3.5 Capacity and voltage determination. After the test specified in 4.5.3.2.1, each cell shall be cycled at 0°C ± 3°C as recommended in the service manual and as follows:

(a) Cycle 1. Each cell shall be charged as recommended in the service manual. Each cell shall then be discharged at 240 A to the corresponding minimum cell voltage.
(b) Cycle 2. Each cell shall be recharged. The cell shall then be discharged at 150 A to the corresponding minimum cell voltage.
(c) Cycle 3. Each cell shall then be recharged at 100 A to the recommended cut-off voltage then immediately discharged at 200 A for 2 hours.

Data on voltage, current, and electrolyte temperature shall be monitored and recorded. The results of the capacity discharge tests at each rate shall be averaged and the estimated standard deviation calculated according to the following formula:

$$s = \left(\frac{\sum_{1}^{n} (x_i - \bar{x})^2}{n-1} \right)^{1/2}$$

Where:

s = Estimated standard deviation.
n = Number of sample cells.
$\bar{x}$ = Average of results of test.
x_i = Result of individual sample cell.

Capacity and voltage of each cell shall meet the requirements of 3.3.1 and the average percent of rated capacity for all cells shall be equal to or greater than 100 percent of rated capacity plus 3.1 s."

PAGE 11

6.2.1(g): Delete "repair parts" and substitute "installation and checkout parts".

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Figure 2: Delete and substitute the attached figure 2.

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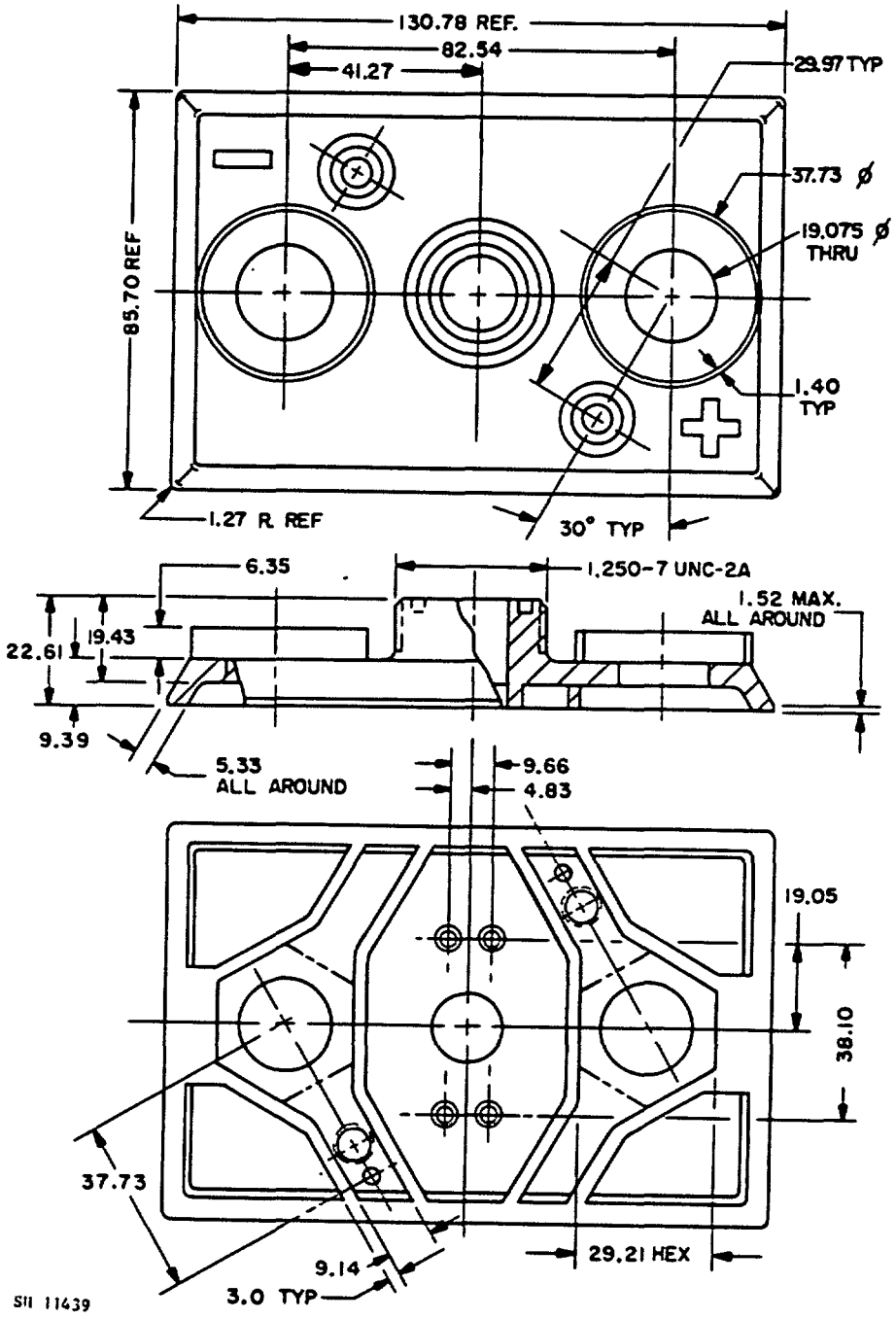
LAST PAGE

DD Form 1426, Standardization Document Improvement Proposal: Delete address and substitute:

"COMMANDER
NAVAL SEA SYSTEMS COMMAND (SEA 55Z3)
DEPARTMENT OF THE NAVY
WASHINGTON, DC 20362"

Preparing activity:
Navy - SH
(Project 6140-N574)

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NOTE:

1. All dimensions in millimeters are nominal. Threaded hardware designation is unified thread standard.
2. Stress relieve (anneal) in air at 165°C ± 3°C for at least 4 hours.

FIGURE 2. Cell cover.

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